O*NET OnLine

Network and Computer Systems Administrators

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Install, configure, and maintain an organization's local area network (LAN), wide area network (WAN), data communications network, operating systems, and physical and virtual servers. Perform system monitoring and verify the integrity and availability of hardware, network, and server resources and systems. Review system and application logs and verify completion of scheduled jobs, including system backups. Analyze network and server resource consumption and control user access. Install and upgrade software and maintain software licenses. May assist in network modeling, analysis, planning, and coordination between network and data communications hardware and software.

Sample of reported job titles: LAN Administrator (Local Area Network Administrator), LAN Specialist (Local Area Network Specialist), Network Administrator (Network Admin), Network Analyst, Network Coordinator, Network Engineer, Network Manager, Network Specialist, Systems Engineer

Summary | Details | Custom | Easy Read | Veterans | Español

Contents

Occupation-Specific Information

Tasks

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Importance	Category	Task
89	Core	Plan, coordinate, and implement network security measures to protect data, software, and hardware.
84	Core	Perform data backups and disaster recovery operations.
78	Core	Diagnose, troubleshoot, and resolve hardware, software, or other network and system problems, and replace defective components when necessary.

Importance	Category	Task
77	Core	Maintain and administer computer networks and related computing environments, including computer hardware, systems software, applications software, and all configurations.
73	Core	Configure, monitor, and maintain email applications or virus protection software.
70	Core	Confer with network users about solutions to existing system problems.
68	Core	Operate master consoles to monitor the performance of computer systems and networks and to coordinate computer network access and use.
66	Core	Maintain logs related to network functions, as well as maintenance and repair records.
64	Core	Monitor network performance to determine whether adjustments are needed and where changes will be needed in the future.
62	Core	Design, configure, and test computer hardware, networking software and operating system software.
61	Core	Maintain an inventory of parts for emergency repairs.
59	Core	Perform routine network startup and shutdown procedures, and maintain control records.
58	Core	Recommend changes to improve systems and network configurations, and determine hardware or software requirements related to such changes.
58	Core	Coordinate with vendors and with company personnel to facilitate purchases.
57	Core	Gather data pertaining to customer needs, and use the information to identify, predict, interpret, and evaluate system and network requirements.
57	Core	Analyze equipment performance records to determine the need for repair or replacement.
56	Core	Train people in computer system use.

Importance	Category	Task
51	Core	Research new technologies by attending seminars, reading trade articles, or taking classes, and implement or recommend the implementation of new technologies.
49	Supplemental	Implement and provide technical support for voice services and equipment, such as private branch exchange, voice mail system, and telecom system.
45	Supplemental	Load computer tapes and disks, and install software and printer paper or forms.

Find occupations related to multiple tasks

Occupational Requirements

Work Activities

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Importance	Work Activity
96	**Working with Computers** — Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.
81	**Getting Information** — Observing, receiving, and otherwise obtaining information from all relevant sources.
80	**Making Decisions and Solving Problems** — Analyzing information and evaluating results to choose the best solution and solve problems.
78	**Analyzing Data or Information** — Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.
78	**Updating and Using Relevant Knowledge** — Keeping up-to-date technically and applying new knowledge to your job.
76	**Processing Information** — Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.

Importance	Work Activity
75	**Documenting/Recording Information** — Entering, transcribing, recording, storing, or maintaining information in written or electronic/magnetic form.
73	**Communicating with Supervisors, Peers, or Subordinates** — Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.
73	**Evaluating Information to Determine Compliance with Standards** — Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.
68	**Identifying Objects, Actions, and Events** — Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
65	**Monitoring Processes, Materials, or Surroundings** — Monitoring and reviewing information from materials, events, or the environment, to detect or assess problems.
65	**Thinking Creatively** — Developing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.
63	**Organizing, Planning, and Prioritizing Work** — Developing specific goals and plans to prioritize, organize, and accomplish your work.
59	**Developing Objectives and Strategies** — Establishing long-range objectives and specifying the strategies and actions to achieve them.
54	**Training and Teaching Others** — Identifying the educational needs of others, developing formal educational or training programs or classes, and teaching or instructing others.
53	**Establishing and Maintaining Interpersonal Relationships** — Developing constructive and cooperative working relationships with others, and maintaining them over time.
53	**Interpreting the Meaning of Information for Others** — Translating or explaining what information means and how it can be used.
50	**Inspecting Equipment, Structures, or Materials** — Inspecting equipment, structures, or materials to identify the cause of errors or other problems or defects.

Importance	Work Activity
50	**Performing Administrative Activities** — Performing day-to-day administrative tasks such as maintaining information files and processing paperwork.
49	**Estimating the Quantifiable Characteristics of Products, Events, or Information** — Estimating sizes, distances, and quantities; or determining time, costs, resources, or materials needed to perform a work activity.
49	**Repairing and Maintaining Electronic Equipment** — Servicing, repairing, calibrating, regulating, fine-tuning, or testing machines, devices, and equipment that operate primarily on the basis of electrical or electronic (not mechanical) principles.
46	**Developing and Building Teams** — Encouraging and building mutual trust, respect, and cooperation among team members.
46	**Judging the Qualities of Objects, Services, or People** — Assessing the value, importance, or quality of things or people.
45	**Communicating with People Outside the Organization** — Communicating with people outside the organization, representing the organization to customers, the public, government, and other external sources. This information can be exchanged in person, in writing, or by telephone or e-mail.
45	**Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment** — Providing documentation, detailed instructions, drawings, or specifications to tell others about how devices, parts, equipment, or structures are to be fabricated, constructed, assembled, modified, maintained, or used.
45	**Providing Consultation and Advice to Others** — Providing guidance and expert advice to management or other groups on technical, systems-, or process-related topics.
45	**Scheduling Work and Activities** — Scheduling events, programs, and activities, as well as the work of others.
43	**Guiding, Directing, and Motivating Subordinates** — Providing guidance and direction to subordinates, including setting performance standards and monitoring performance.
39	**Coaching and Developing Others** — Identifying the developmental needs of others and coaching, mentoring, or otherwise helping others to improve their knowledge or skills.

Importance	Work Activity
34	**Coordinating the Work and Activities of Others** — Getting members of a group to work together to accomplish tasks.
34	**Handling and Moving Objects** — Using hands and arms in handling, installing, positioning, and moving materials, and manipulating things.
31	**Monitoring and Controlling Resources** — Monitoring and controlling resources and overseeing the spending of money.
26	**Performing General Physical Activities** — Performing general physical activities includes doing activities that require considerable use of your arms and legs and moving your whole body, such as climbing, lifting, balancing, walking, stooping, and handling materials.
25	**Resolving Conflicts and Negotiating with Others** — Handling complaints, settling disputes, and resolving grievances and conflicts, or otherwise negotiating with others.
18	**Assisting and Caring for Others** — Providing personal assistance, medical attention, emotional support, or other personal care to others such as coworkers, customers, or patients.
18	**Repairing and Maintaining Mechanical Equipment** — Servicing, repairing, adjusting, and testing machines, devices, moving parts, and equipment that operate primarily on the basis of mechanical (not electronic) principles.
17	**Controlling Machines and Processes** — Using either control mechanisms or direct physical activity to operate machines or processes (not including computers or vehicles).
17	**Operating Vehicles, Mechanized Devices, or Equipment** — Running, maneuvering, navigating, or driving vehicles or mechanized equipment, such as forklifts, passenger vehicles, aircraft, or watercraft.
17	**Staffing Organizational Units** — Recruiting, interviewing, selecting, hiring, and promoting employees in an organization.
14	**Selling or Influencing Others** — Convincing others to buy merchandise/goods or to otherwise change their minds or actions.
13	**Performing for or Working Directly with the Public** — Performing for people or dealing directly with the public. This includes serving customers in restaurants and stores, and receiving clients or guests.

Detailed Work Activities

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- Implement security measures for computer or information systems.
- Develop computer or information security policies or procedures.
- Create electronic data backup to prevent loss of information.
- Resolve computer network problems.
- Resolve computer software problems.
- Troubleshoot issues with computer applications or systems.
- Maintain computer networks to enhance performance and user access.
- Provide technical support for computer network issues.
- Collaborate with others to resolve information technology issues.
- Monitor the performance of computer networks.
- Document network-related activities or tasks.
- Design integrated computer systems.
- Test computer hardware performance.
- Test software performance.
- Maintain the inventory of equipment.
- Document operational activities.
- Analyze project data to determine specifications or requirements.
- Identify information technology project resource requirements.
- Recommend changes to improve computer or information systems.
- Coordinate resource procurement activities.
- Analyze data to identify or resolve operational problems.
- Collect data about customer needs.
- Train others in computer interface or software use.
- Conduct research to gain information about products or processes.
- Update knowledge about emerging industry or technology trends.
- Install computer hardware.

Find occupations related to multiple detailed work activities

Work Context

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- **E-Mail** — How frequently does your job require you to use E-mail?
  - 100% Every day
- **Telephone Conversations** — How often do you have telephone conversations in this job?
  - 77% Every day
  - 20% Once a week or more but not every day
- **Indoors, Environmentally Controlled** — How often does this job require working indoors in an environmentally controlled environment (like a warehouse with air conditioning)?
  - 80% Every day
- **Face-to-Face Discussions with Individuals and Within Teams** — How frequently does your job require face-to-face discussions with individuals and within teams?
  - 50% Every day
  - 33% Once a week or more but not every day
  - 13% Once a month or more but not every week
- **Spend Time Sitting** — How much does this job require sitting?
  - 50% Continually or almost continually
  - 33% More than half the time
- **Work With or Contribute to a Work Group or Team** — How important is it to work with or contribute to a work group or team in this job?
  - 52% Extremely important
  - 28% Very important
  - 14% Important
- **Contact With Others** — How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?
  - 50% Constant contact with others
  - 23% Contact with others most of the time
  - 20% Contact with others about half the time
- **Determine Tasks, Priorities and Goals** — How much freedom does the worker have in determining the tasks, priorities, or goals of the job?
  - 30% A lot of freedom
  - 50% Some freedom
  - 13% Limited freedom
- **Freedom to Make Decisions** — How much decision making freedom, without supervision, does the job offer?
  - 30% A lot of freedom
  - 37% Some freedom

30% Limited freedom

Importance of Being Exact or Accurate — How important is being very exact or highly accurate in performing this job?

30% Extremely important

37% Very important

27% Important

Duration of Typical Work Week — Number of hours typically worked in one week.

40% More than 40 hours

57% 40 hours

Consequence of Error — How serious would the result usually be if the worker made a mistake that was not easily correctable?

23% Extremely serious

43% Very serious

17% Serious

17% Fairly serious

Impact of Decisions on Co-workers or Company Results — What results do your decisions usually have on other people or the image or reputation or financial resources of your employer?

27% Very important results

30% Important results

33% Moderate results

Time Pressure — How often does this job require the worker to meet strict deadlines?

17% Every day

33% Once a week or more but not every day

37% Once a month or more but not every week

Frequency of Decision Making — How often is the worker required to make decisions that affect other people, the financial resources, and/or the image and reputation of the organization?

33% Every day

20% Once a week or more but not every day

33% Once a year or more but not every month

Level of Competition — To what extent does this job require the worker to compete or to be aware of competitive pressures?

13% Extremely competitive

33% Highly competitive

33% Moderately competitive

17% Slightly competitive

- **Coordinate or Lead Others in Accomplishing Work Activities** — How important is it to coordinate or lead others (not as a supervisor or team leader) in accomplishing work activities in this job?
  - 20% Extremely important
  - 27% Very important
  - 27% Important
  - 13% Fairly important
  - 13% Not important at all
- **Importance of Repeating Same Tasks** — How important are continuous, repetitive, physical activities (like key entry) or mental activities (like checking entries in a ledger) to performing this job?
  - 13% Extremely important
  - 27% Very important
  - 23% Important
  - 23% Fairly important
  - 13% Not important at all
- **Work Outcomes and Results of Other Workers** — How responsible is the worker for work outcomes and results of other workers?
  - 30% High responsibility
  - 30% Moderate responsibility
  - 27% Limited responsibility
- **Degree of Automation** — How automated is the job?
  - 27% Highly automated
  - 50% Moderately automated
  - 20% Slightly automated
- **Physical Proximity** — To what extent does this job require the worker to perform job tasks physically close to other people?
  - 60% Slightly close (e.g., shared office)
  - 20% I work with others but not closely (e.g., private office)
- **Written Letters and Memos** — How frequently does your job require written letters and memos?
  - 17% Every day
  - 13% Once a week or more but not every day
  - 30% Once a month or more but not every week
  - 17% Once a year or more but not every month
  - 23% Never
- **Spend Time Using Your Hands to Handle, Control, or Feel Objects, Tools, or Controls** — How much does this job require using your hands to handle, control, or feel objects, tools or controls?

- 20% Continually or almost continually
- 47% Less than half the time
- 13% Never

Deal With External Customers or the Public in General — How important is it to deal with external customers (as in retail sales) or the public in general (as in police work) in this job?

- 13% Extremely important
- 17% Very important
- 40% Fairly important
- 20% Not important at all

Dealing With Unpleasant, Angry, or Discourteous People — How frequently does the worker have to deal with unpleasant, angry, or discourteous individuals as part of the job requirements?

- 13% Once a week or more but not every day
- 37% Once a month or more but not every week
- 33% Once a year or more but not every month
- 17% Never

Exposed to Sounds, Noise Levels that are Distracting or Uncomfortable — How often does this job require working exposed to sounds and noise levels that are distracting or uncomfortable?

- 13% Once a week or more but not every day
- 23% Once a month or more but not every week
- 17% Once a year or more but not every month
- 37% Never

Spend Time Making Repetitive Motions — How much does this job require making repetitive motions?

- 20% About half the time
- 33% Less than half the time
- 27% Never

Spend Time Standing — How much does this job require standing?

- 14% About half the time
- 59% Less than half the time

Health and Safety of Other Workers — How much responsibility is there for the health and safety of others in this job?

- 13% High responsibility
- 37% Limited responsibility
- 30% No responsibility

Public Speaking — How frequently does your job require public speaking (one speaker with an audience)?

17% Once a month or more but not every week

43% Once a year or more but not every month

23% Never

Conflict Situations — How frequently are there conflict situations the employee has to face in this job?

33% Once a month or more but not every week

40% Once a year or more but not every month

20% Never

Spend Time Walking or Running — How much does this job require walking or running?

20% About half the time

50% Less than half the time

23% Never

Spend Time Kneeling, Crouching, Stooping, or Crawling — How much does this job require kneeling, crouching, stooping or crawling?

63% Less than half the time

23% Never

Indoors, Not Environmentally Controlled — How often does this job require working in an environment that is not environmentally controlled (like a warehouse without air conditioning)?

23% Once a year or more but not every month

53% Never

Spend Time Bending or Twisting Your Body — How much does this job require bending or twisting your body?

50% Less than half the time

37% Never

Wear Common Protective or Safety Equipment such as Safety Shoes, Glasses, Gloves, Hearing Protection, Hard Hats, or Life Jackets — How often does this job require wearing common protective or safety equipment such as safety shoes, glasses, gloves, hearing protection, hard hats or life-jackets?

17% Once a year or more but not every month

60% Never

Exposed to Cramped Work Space, Awkward Positions — How often does this job require working in cramped work spaces that requires getting into awkward positions?

20% Once a month or more but not every week

27% Once a year or more but not every month

50% Never

In an Enclosed Vehicle or Operate Enclosed Equipment — How often does this job require working in a closed vehicle or operate enclosed equipment (like a car)?

13% Every day

73% Never

Pace Determined by Speed of Equipment — How important is it to this job that the pace is determined by the speed of equipment or machinery? (This does not refer to keeping busy at all times on this job.)

17% Important

67% Not important at all

Spend Time Climbing Ladders, Scaffolds, or Poles — How much does this job require climbing ladders, scaffolds, poles, etc.?

37% Less than half the time

53% Never

Exposed to High Places — How often does this job require exposure to high places?

13% Once a month or more but not every week

70% Never

Outdoors, Exposed to All Weather Conditions — How often does this job require working outdoors, exposed to all weather conditions?

20% Once a year or more but not every month

67% Never

Exposed to Extremely Bright or Inadequate Lighting Conditions — How often does this job require working in extremely bright or inadequate lighting conditions?

13% Once a month or more but not every week

17% Once a year or more but not every month

67% Never

Exposed to Very Hot or Cold Temperatures — How often does this job require working in very hot (above 90 F degrees) or very cold (below 32 F degrees) temperatures?

17% Once a year or more but not every month

70% Never

Exposed to Minor Burns, Cuts, Bites, or Stings — How often does this job require exposure to minor burns, cuts, bites, or stings?

24% Once a year or more but not every month

66% Never

Spend Time Keeping or Regaining Balance — How much does this job require keeping or regaining your balance?

20% Less than half the time

73% Never

Outdoors, Under Cover — How often does this job require working outdoors, under cover (like in an open shed)?

14% Once a year or more but not every month

76% Never

Exposed to Hazardous Conditions — How often does this job require exposure to hazardous conditions?

83% Never

Exposed to Contaminants — How often does this job require working exposed to contaminants (such as pollutants, gases, dust or odors)?

86% Never

Work Schedules — How regular are the work schedules for this job?

13% Irregular (changes with weather conditions, production demands, or contract duration)

87% Regular (established routine, set schedule)

In an Open Vehicle or Operating Equipment — How often does this job require working in an open vehicle or operating equipment (like a tractor)?

93% Never

Wear Specialized Protective or Safety Equipment such as Breathing Apparatus, Safety Harness, Full Protection Suits, or Radiation Protection — How often does this job require wearing specialized protective or safety equipment such as breathing apparatus, safety harness, full protection suits, or radiation protection?

90% Never

Exposed to Hazardous Equipment — How often does this job require exposure to hazardous equipment?

93% Never

Exposed to Whole Body Vibration — How often does this job require exposure to whole body vibration (like operating a jackhammer or earth moving equipment)?

93% Never

Dealing with Violent or Physically Aggressive People — How frequently does this job require the worker to deal with physical aggression of violent individuals?

93% Never

Exposed to Disease or Infections — How often does this job require exposure to disease/infections?

97% Never

Exposed to Radiation — How often does this job require exposure to radiation?

100% Never

Experience Requirements

Job Zone

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Title	Job Zone Three: Medium Preparation Needed
Education	Most occupations in this zone require training in vocational schools, related on-the-job experience, or an associate's degree.
Related Experience	Previous work-related skill, knowledge, or experience is required for these occupations. For example, an electrician must have completed three or four years of apprenticeship or several years of vocational training, and often must have passed a licensing exam, in order to perform the job.
Job Training	Employees in these occupations usually need one or two years of training involving both on-the-job experience and informal training with experienced workers. A recognized apprenticeship program may be associated with these occupations.
Job Zone Examples	These occupations usually involve using communication and organizational skills to coordinate, supervise, manage, or train others to accomplish goals. Examples include hydroelectric production managers, desktop publishers, electricians, agricultural technicians, barbers, court reporters and simultaneous captioners, and medical assistants.
SVP Range	(6.0 to < 7.0)

Training & Credentials

State training Select a State Go

Local training ZIP Code Go

Certifications Find certifications nationwide

Apprenticeship Opportunities

All 7 displayed

Example apprenticeship titles for this occupation:

- Information Assurance Specialist
- Information Technology Specialist
- It Systems Administrator
- Network and Computer Systems Administrator (Alternate Title: It Specialist)

- Network Support Technician
- Systems Administrator
- Systems Administrator (Nof)

Specific title(s) listed above are vetted by industry and approved by the U.S. Department of Labor for use in a Registered Apprenticeship Program.

Start your career and build your skillset. Visit **Apprenticeship.gov** to learn about opportunities related to this occupation.

Worker Requirements

Software Skills

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- **Access software** — Access management software; Citrix cloud computing software; Mac HelpMate; Remote desktop control software; 1 more
- **Administration software** — Cisco Systems CiscoWorks; Hewlett-Packard HP Network Node Manager; Network management software; Network shutdown software; 5 more
- **Analytical or scientific software** — Root cause analysis software; SAS ; StataCorp Stata
- **Application server software** — Docker ; GitHub ; Oracle WebLogic Server; Red Hat OpenShift ; 2 more
- **Authentication server software** — Password management software
- **Backup or archival software** — System and data disaster recovery software; Veritas NetBackup
- **Business intelligence and data analysis software** — Apache Spark ; IBM Cognos Impromptu; Oracle Business Intelligence Enterprise Edition; Tableau ; 2 more
- **Cloud-based management software** — Amazon Web Services AWS CloudFormation ; IBM WebSphere; Splunk Enterprise
- **Cloud-based protection or security software** — SolarWinds
- **Clustering software** — VMware
- **Communications server software** — IBM Domino
- **Compiler and decompiler software** — Command interpreters
- **Computer aided design CAD software** — Computer aided design and drafting CADD software; Dassault Systemes CATIA
- **Computer based training software** — Moodle
- **Computer imaging software** — Symantec Ghost Solution Suite
- **Configuration management software** — Chef ; Patch and update management software; Perforce Helix software; Puppet ; 3 more
- **Content workflow software** — Atlassian JIRA

- **Customer relationship management CRM software** — Blackbaud The Raiser's Edge
- **Data base management system software** — Amazon DynamoDB; Apache Hive; Elasticsearch; Oracle PL/SQL; 9 more
- **Data base reporting software** — Microsoft SQL Server Reporting Services SSRS; SAP Crystal Reports
- **Data base user interface and query software** — Amazon Elastic Compute Cloud EC2; Blackboard software; ServiceNow; Transact-SQL; 9 more
- **Desktop communications software** — ParentSquare; Secure shell SSH software; Skype
- **Desktop publishing software** — Microsoft Publisher
- **Development environment software** — Apache Kafka; Apache Maven; Go; Microsoft PowerShell; 14 more
- **Document management software** — Adobe Acrobat; Microsoft SharePoint
- **Electronic mail software** — IBM Notes; Microsoft Exchange; Microsoft Outlook
- **Enterprise application integration software** — Extensible markup language XML; IBM InfoSphere DataStage; Microsoft SQL Server Integration Services SSIS; Systems integration software; 3 more
- **Enterprise resource planning ERP software** — Microsoft Dynamics; Microsoft Dynamics GP; NetSuite ERP; SAP software; 7 more
- **Enterprise system management software** — IBM Power Systems software
- **Expert system software** — Ansible software
- **File versioning software** — Apache Subversion SVN; Git
- **Financial analysis software** — Delphi Technology; Oracle E-Business Suite Financials
- **Gateway software** — Microsoft Windows Terminal Services Access Manager
- **Geographic information system** — ESRI ArcGIS software; Geographic information system GIS software
- **Graphical user interface development software** — TKSoftware RCM software
- **Helpdesk or call center software** — Help desk software
- **Human resources software** — Human resource management software HRMS; Oracle Taleo
- **Industrial control software** — Supervisory control and data acquisition SCADA software
- **Internet directory services software** — Active directory software; Domain name system DNS; Microsoft Active Directory; Network addressable storage NAS software; 1 more
- **Internet protocol IP multimedia subsystem software** — Open source routing protocols OSPF; Voice over internet protocol VoIP system software
- **Medical software** — Epic Systems
- **Metadata management software** — Quest Erwin Data Modeler
- **Network monitoring software** — Nagios; Quest Foglight; Remote monitoring software; Wireshark; 23 more
- **Network operating system enhancement software** — Management information base MIB software; Network, server and operating system optimization software; Operating system process control software
- **Network security and virtual private network VPN equipment software** — Firewall software; Network intrusion detection software; Virtual private networking VPN software

- **Network security or virtual private network VPN management software** — Intrusion prevention system IPS; Network and system vulnerability assessment software; OpenService Open NerveCenter; Security incident management software; 1 more
- **Object or component oriented development software** — C#; jQuery; Perl; Swift; 7 more
- **Object oriented data base management software** — Hibernate ORM; PostgreSQL
- **Office suite software** — Microsoft Office software
- **Operating system software** — Bash; Microsoft Windows Server; Red Hat Enterprise Linux; UNIX Shell; 10 more
- **Pattern design software** — Diagramming software
- **Platform interconnectivity software** — Virtual network computing VNC software
- **Portal server software** — Apache HTTP Server
- **Presentation software** — Microsoft PowerPoint
- **Process mapping and design software** — Microsoft Visio
- **Program testing software** — Computer system diagnostics software; Hewlett Packard LoadRunner
- **Project management software** — Atlassian Confluence; Microsoft Project; Microsoft Teams; Oracle Primavera Enterprise Project Portfolio Management
- **Requirements analysis and system architecture software** — Requirements management software
- **Sales and marketing software** — Marketo Marketing Automation
- **Spreadsheet software** — Microsoft Excel
- **Storage networking software** — Amazon Simple Storage Service S3; Storage area network SAN software
- **Switch or router software** — Border Gateway Protocol BGP
- **Transaction security and virus protection software** — Honeypot; McAfee; NortonLifeLock cybersecurity software; Root kit detection software; 3 more
- **Transaction server software** — Customer information control system CICS
- **Video conferencing software** — Cisco Webex
- **Web page creation and editing software** — Adobe Dreamweaver; Google Sites
- **Web platform development software** — Apache Tomcat; Django; Microsoft ASP.NET; Node.js; 15 more

Hot Technologies are requirements most frequently included across all employer job postings.
See all 95 Hot Technologies for this occupation.

In Demand skills are frequently included in employer job postings for this occupation.
See all 18 In Demand skills for this occupation.

Essential Skills

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Importance	Essential Skill
75	**Active Listening** — Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
72	**Critical Thinking** — Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions, or approaches to problems.
72	**Monitoring** — Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
72	**Reading Comprehension** — Understanding written sentences and paragraphs in work-related documents.
56	**Speaking** — Talking to others to convey information effectively.
53	**Active Learning** — Understanding the implications of new information for both current and future problem-solving and decision-making.
50	**Learning Strategies** — Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
50	**Writing** — Communicating effectively in writing as appropriate for the needs of the audience.
44	**Mathematics** — Using mathematics to solve problems.
28	**Science** — Using scientific rules and methods to solve problems.

Transferable Skills

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Importance	Transferable Skill
75	**Judgment and Decision Making** — Considering the relative costs and benefits of potential actions to choose the most appropriate one.
75	**Systems Analysis** — Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.

Importance	Transferable Skill
75	**Systems Evaluation** — Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
75	**Troubleshooting** — Determining causes of operating errors and deciding what to do about it.
72	**Complex Problem Solving** — Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.
63	**Operations Monitoring** — Watching gauges, dials, or other indicators to make sure a machine is working properly.
63	**Programming** — Writing computer programs for various purposes.
60	**Quality Control Analysis** — Conducting tests and inspections of products, services, or processes to evaluate quality or performance.
56	**Time Management** — Managing one's own time and the time of others.
53	**Equipment Maintenance** — Performing routine maintenance on equipment and determining when and what kind of maintenance is needed.
53	**Operation and Control** — Controlling operations of equipment or systems.
53	**Repairing** — Repairing machines or systems using the needed tools.
50	**Coordination** — Adjusting actions in relation to others' actions.
50	**Instructing** — Teaching others how to do something.
50	**Operations Analysis** — Analyzing needs and product requirements to create a design.
47	**Equipment Selection** — Determining the kind of tools and equipment needed to do a job.
47	**Management of Personnel Resources** — Motivating, developing, and directing people as they work, identifying the best people for the job.
47	**Social Perceptiveness** — Being aware of others' reactions and understanding why they react as they do.
44	**Service Orientation** — Actively looking for ways to help people.

Importance	Transferable Skill
44	**Technology Design** — Generating or adapting equipment and technology to serve user needs.
41	**Installation** — Installing equipment, machines, wiring, or programs to meet specifications.
38	**Persuasion** — Persuading others to change their minds or behavior.
35	**Negotiation** — Bringing others together and trying to reconcile differences.
31	**Management of Material Resources** — Obtaining and seeing to the appropriate use of equipment, facilities, and materials needed to do certain work.
19	**Management of Financial Resources** — Determining how money will be spent to get the work done, and accounting for these expenditures.

Knowledge

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Importance	Knowledge
91	**Computers and Electronics** — Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
64	**Telecommunications** — Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.
59	**Customer and Personal Service** — Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.
59	**English Language** — Knowledge of the structure and content of the English language including the meaning and spelling of words, and rules of composition and grammar.
54	**Administration and Management** — Knowledge of business and management principles involved in strategic planning, resource allocation, human resources

Importance	Knowledge
	modeling, leadership technique, production methods, and coordination of people and resources.
53	**Administrative** — Knowledge of administrative and office procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and workplace terminology.
49	**Engineering and Technology** — Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.
36	**Design** — Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.
35	**Mathematics** — Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.
34	**Public Safety and Security** — Knowledge of relevant equipment, policies, procedures, and strategies to promote effective local, state, or national security operations for the protection of people, data, property, and institutions.
33	**Communications and Media** — Knowledge of media production, communication, and dissemination techniques and methods. This includes alternative ways to inform and entertain via written, oral, and visual media.
28	**Education and Training** — Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.
27	**Law and Government** — Knowledge of laws, legal codes, court procedures, precedents, government regulations, executive orders, agency rules, and the democratic political process.
23	**Personnel and Human Resources** — Knowledge of principles and procedures for personnel recruitment, selection, training, compensation and benefits, labor relations and negotiation, and personnel information systems.
18	**Mechanical** — Knowledge of machines and tools, including their designs, uses, repair, and maintenance.
18	**Production and Processing** — Knowledge of raw materials, production processes, quality control, costs, and other techniques for maximizing the effective manufacture and distribution of goods.

Importance	Knowledge
15	**Economics and Accounting** — Knowledge of economic and accounting principles and practices, the financial markets, banking, and the analysis and reporting of financial data.
15	**Psychology** — Knowledge of human behavior and performance; individual differences in ability, personality, and interests; learning and motivation; psychological research methods; and the assessment and treatment of behavioral and affective disorders.
13	**Sales and Marketing** — Knowledge of principles and methods for showing, promoting, and selling products or services. This includes marketing strategy and tactics, product demonstration, sales techniques, and sales control systems.
12	**Geography** — Knowledge of principles and methods for describing the features of land, sea, and air masses, including their physical characteristics, locations, interrelationships, and distribution of plant, animal, and human life.
11	**Transportation** — Knowledge of principles and methods for moving people or goods by air, rail, sea, or road, including the relative costs and benefits.
8	**Building and Construction** — Knowledge of materials, methods, and the tools involved in the construction or repair of houses, buildings, or other structures such as highways and roads.
8	**Physics** — Knowledge and prediction of physical principles, laws, their interrelationships, and applications to understanding fluid, material, and atmospheric dynamics, and mechanical, electrical, atomic and sub-atomic structures and processes.
8	**Sociology and Anthropology** — Knowledge of group behavior and dynamics, societal trends and influences, human migrations, ethnicity, cultures, and their history and origins.
5	**Foreign Language** — Knowledge of the structure and content of a foreign (non-English) language including the meaning and spelling of words, rules of composition and grammar, and pronunciation.
3	**Chemistry** — Knowledge of the chemical composition, structure, and properties of substances and of the chemical processes and transformations that they undergo. This includes uses of chemicals and their interactions, danger signs, production techniques, and disposal methods.
3	**Food Production** — Knowledge of techniques and equipment for planting, growing, and harvesting food products (both plant and animal) for consumption,

Importance	Knowledge
	including storage/handling techniques.
3	**Therapy and Counseling** — Knowledge of principles, methods, and procedures for diagnosis, treatment, and rehabilitation of physical and mental dysfunctions, and for career counseling and guidance.
2	**Fine Arts** — Knowledge of the theory and techniques required to compose, produce, and perform works of music, dance, visual arts, drama, and sculpture.
1	**Biology** — Knowledge of plant and animal organisms, their tissues, cells, functions, interdependencies, and interactions with each other and the environment.
1	**Medicine and Dentistry** — Knowledge of the information and techniques needed to diagnose and treat human injuries, diseases, and deformities. This includes symptoms, treatment alternatives, drug properties and interactions, and preventive health-care measures.
0	**History and Archeology** — Knowledge of historical events and their causes, indicators, and effects on civilizations and cultures.
0	**Philosophy and Theology** — Knowledge of different philosophical systems and religions. This includes their basic principles, values, ethics, ways of thinking, customs, practices, and their impact on human culture.

Education

How much education does a new hire need to perform a job in this occupation? Respondents said:

- 30% Associate's degree or other 2-year degree required
- 27% Bachelor's degree required
- 20% Post-secondary certificate required

Worker Characteristics

Abilities

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Importance	Ability
75	**Deductive Reasoning** — The ability to apply general rules to specific problems to produce answers that make sense.
75	**Inductive Reasoning** — The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).
75	**Information Ordering** — The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).
75	**Problem Sensitivity** — The ability to tell when something is wrong or is likely to go wrong. It does not involve solving the problem, only recognizing that there is a problem.
72	**Near Vision** — The ability to see details at close range (within a few feet of the observer).
72	**Oral Comprehension** — The ability to listen to and understand information and ideas presented through spoken words and sentences.
72	**Oral Expression** — The ability to communicate information and ideas in speaking so others will understand.
72	**Written Comprehension** — The ability to read and understand information and ideas presented in writing.
56	**Fluency of Ideas** — The ability to come up with a number of ideas about a topic (the number of ideas is important, not their quality, correctness, or creativity).
53	**Flexibility of Closure** — The ability to identify or detect a known pattern (a figure, object, word, or sound) that is hidden in other distracting material.
53	**Originality** — The ability to come up with unusual or clever ideas about a given topic or situation, or to develop creative ways to solve a problem.
53	**Visualization** — The ability to imagine how something will look after it is moved around or when its parts are moved or rearranged.
53	**Written Expression** — The ability to communicate information and ideas in writing so others will understand.
50	**Category Flexibility** — The ability to generate or use different sets of rules for combining or grouping things in different ways.

Importance	Ability
50	**Perceptual Speed** — The ability to quickly and accurately compare similarities and differences among sets of letters, numbers, objects, pictures, or patterns. The things to be compared may be presented at the same time or one after the other. This ability also includes comparing a presented object with a remembered object.
50	**Selective Attention** — The ability to concentrate on a task over a period of time without being distracted.
50	**Speech Clarity** — The ability to speak clearly so others can understand you.
50	**Speech Recognition** — The ability to identify and understand the speech of another person.
47	**Speed of Closure** — The ability to quickly make sense of, combine, and organize information into meaningful patterns.
47	**Visual Color Discrimination** — The ability to match or detect differences between colors, including shades of color and brightness.
44	**Arm-Hand Steadiness** — The ability to keep your hand and arm steady while moving your arm or while holding your arm and hand in one position.
44	**Far Vision** — The ability to see details at a distance.
44	**Manual Dexterity** — The ability to quickly move your hand, your hand together with your arm, or your two hands to grasp, manipulate, or assemble objects.
44	**Mathematical Reasoning** — The ability to choose the right mathematical methods or formulas to solve a problem.
44	**Memorization** — The ability to remember information such as words, numbers, pictures, and procedures.
44	**Number Facility** — The ability to add, subtract, multiply, or divide quickly and correctly.
41	**Control Precision** — The ability to quickly and repeatedly adjust the controls of a machine or a vehicle to exact positions.
41	**Finger Dexterity** — The ability to make precisely coordinated movements of the fingers of one or both hands to grasp, manipulate, or assemble very small objects.

Importance	Ability
41	**Time Sharing** — The ability to shift back and forth between two or more activities or sources of information (such as speech, sounds, touch, or other sources).
25	**Auditory Attention** — The ability to focus on a single source of sound in the presence of other distracting sounds.
25	**Wrist-Finger Speed** — The ability to make fast, simple, repeated movements of the fingers, hands, and wrists.
22	**Depth Perception** — The ability to judge which of several objects is closer or farther away from you, or to judge the distance between you and an object.
22	**Hearing Sensitivity** — The ability to detect or tell the differences between sounds that vary in pitch and loudness.
22	**Trunk Strength** — The ability to use your abdominal and lower back muscles to support part of the body repeatedly or continuously over time without "giving out" or fatiguing.
19	**Extent Flexibility** — The ability to bend, stretch, twist, or reach with your body, arms, and/or legs.
19	**Multilimb Coordination** — The ability to coordinate two or more limbs (for example, two arms, two legs, or one leg and one arm) while sitting, standing, or lying down. It does not involve performing the activities while the whole body is in motion.
19	**Reaction Time** — The ability to quickly respond (with the hand, finger, or foot) to a signal (sound, light, picture) when it appears.
16	**Rate Control** — The ability to time your movements or the movement of a piece of equipment in anticipation of changes in the speed and/or direction of a moving object or scene.
16	**Response Orientation** — The ability to choose quickly between two or more movements in response to two or more different signals (lights, sounds, pictures). It includes the speed with which the correct response is started with the hand, foot, or other body part.
3	**Spatial Orientation** — The ability to know your location in relation to the environment or to know where other objects are in relation to you.

Importance	Ability
0	**Dynamic Flexibility** — The ability to quickly and repeatedly bend, stretch, twist, or reach out with your body, arms, and/or legs.
0	**Dynamic Strength** — The ability to exert muscle force repeatedly or continuously over time. This involves muscular endurance and resistance to muscle fatigue.
0	**Explosive Strength** — The ability to use short bursts of muscle force to propel oneself (as in jumping or sprinting), or to throw an object.
0	**Glare Sensitivity** — The ability to see objects in the presence of a glare or bright lighting.
0	**Gross Body Coordination** — The ability to coordinate the movement of your arms, legs, and torso together when the whole body is in motion.
0	**Gross Body Equilibrium** — The ability to keep or regain your body balance or stay upright when in an unstable position.
0	**Night Vision** — The ability to see under low-light conditions.
0	**Peripheral Vision** — The ability to see objects or movement of objects to one's side when the eyes are looking ahead.
0	**Sound Localization** — The ability to tell the direction from which a sound originated.
0	**Speed of Limb Movement** — The ability to quickly move the arms and legs.
0	**Stamina** — The ability to exert yourself physically over long periods of time without getting winded or out of breath.
0	**Static Strength** — The ability to exert maximum muscle force to lift, push, pull, or carry objects.

Career Interest Types

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Occupational Interest	Career Interest Type
86	**Conventional** — Work involves following procedures and regulations to organize information or data, typically in a business setting. Conventional occupations are often associated with office work, accounting, mathematics/statistics, information technology, finance, or human resources.
53	**Realistic** — Work involves designing, building, or repairing of equipment, materials, or structures, engaging in physical activity, or working outdoors. Realistic occupations are often associated with engineering, mechanics and electronics, construction, woodworking, transportation, machine operation, agriculture, animal services, physical or manual labor, athletics, or protective services.
51	**Investigative** — Work involves studying and researching non-living objects, living organisms, disease or other forms of impairment, or human behavior. Investigative occupations are often associated with physical, life, medical, or social sciences, and can be found in the fields of humanities, mathematics/statistics, information technology, or health care service.
41	**Enterprising** — Work involves managing, negotiating, marketing, or selling, typically in a business setting, or leading or advising people in political and legal situations. Enterprising occupations are often associated with business initiatives, sales, marketing/advertising, finance, management/administration, professional advising, public speaking, politics, or law.
20	**Social** — Work involves helping, teaching, advising, assisting, or providing service to others. Social occupations are often associated with social, health care, personal service, teaching/education, or religious activities.
0	**Artistic** — Work involves creating original visual artwork, performances, written works, food, or music for a variety of media, or applying artistic principles to the design of various objects and materials. Artistic occupations are often associated with visual arts, applied arts and design, performing arts, music, creative writing, media, or culinary art.

Specific Interest Areas

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Occupational Interest	Specific Interest Area
93	**Information Technology** — Work involves designing, developing, testing, and maintaining computer software, hardware, networks, and systems.
49	**Mechanics/Electronics** — Work involves maintaining and repairing of machinery, such as automotive engines, electronic equipment, computers, and communication systems.
36	**Engineering** — Work involves applying science and technology to the design, building, testing, and use of electrical and electronic components, mechanical devices and machines, automotive, marine, and aerospace equipment and vehicles, materials, or structures.
33	**Management/Administration** — Work involves planning, directing, or coordinating the daily operations of an organization, and overseeing personnel, business, or administrative services.
24	**Mathematics/Statistics** — Work involves developing and applying mathematics and statistical theory and methods.
16	**Office Work** — Work involves clerical and administrative tasks for an organization such as answering phones, bookkeeping, updating records, scheduling appointments or meetings, and editing of documents and files.
11	**Teaching/Education** — Work involves teaching through lectures, discussions, and demonstrations in one or more subjects.
10	**Accounting** — Work involves examining, analyzing, and interpreting accounting records for financial statements, or maintaining, auditing, or evaluating records and accounts.
10	**Human Resources** — Work involves recruiting, selecting, and training job applicants, maintaining employee personnel records, administering compensation and employee-benefit programs, and conducting employee performance reviews.
10	**Personal Service** — Work involves providing personal services to individuals in need of non-medical assistance such as helping arrange or plan for travel, meals, or events.
10	**Public Speaking** — Work involves speaking before an audience to educate, entertain, or influence the listeners.

Occupational Interest	Specific Interest Area
9	**Business Initiatives** — Work involves creating strategic plans or taking actions to expand an organization's business or market share.
8	**Protective Service** — Work involves maintaining order and protecting life and property by conducting investigations and enforcing laws and regulations to prevent or solve crimes.
7	**Finance** — Work involves directing and managing financial activities, such as planning and investments for an individual or organization to generate income.
6	**Physical Science** — Work involves studying and researching non-living objects, materials, and phenomena through scientific disciplines such as physics, chemistry, astronomy, meteorology, or geology.
6	**Professional Advising** — Work involves providing expert advice, coaching, and counseling on topics such as careers, educational opportunities, and educational and professional development.
5	**Physical/Manual Labor** — Work involves manually unloading and moving freight, stock, luggage, or other materials, or performing other physical labor.
5	**Sales** — Work involves selling goods or services.
4	**Law** — Work involves studying and applying legal knowledge.
4	**Social Science** — Work involves studying and researching human behavior, society, politics, and social relationships through scientific disciplines such as psychology, sociology, economics, or political science.
3	**Health Care Service** — Work involves providing health care services to individuals to include the diagnosis, prevention, or treatment of diseases and other forms of illness or impairment.
3	**Marketing/Advertising** — Work involves promoting or creating a public image for products, services, or organizations to attract interest, engagement, or sales.
3	**Media** — Work involves producing, developing, writing, directing, or presenting information through print, radio, television, or digital media.
2	**Creative Writing** — Work involves developing creative written works, such as novels, scripts, song lyrics, poetry, stories, or advertisements.

Occupational Interest	Specific Interest Area
2	**Social Service** — Work involves providing social services and assistance to improve the well-being and social and psychological functioning of children and adults.
1	**Construction/Woodwork** — Work involves constructing, installing, or repairing structures and fixtures made of wood.
1	**Humanities** — Work involves studying and researching aspects of human history, culture, and language through disciplines such as history, philosophy, the language arts, or area/ethnic studies.
1	**Medical Science** — Work involves studying and researching the causes, nature, effects, and treatments of disease or other forms of impairment or illness.
1	**Politics** — Work involves engaging in political activities to influence government policy.
1	**Transportation/Machine Operation** — Work involves driving vehicles, such as vans, trucks, trains, and buses to transport people or materials. Work can also involve operating heavy equipment, such as cranes, forklifts, bulldozers, and backhoes to move materials or earth.
0	**Agriculture** — Work involves planting, cultivating, and harvesting crops, plants, or trees, or raising farm animals. Work can also involve preparing plant and animal products for distribution to consumers, or protecting natural resources to maximize their use in farming.
0	**Animal Service** — Work involves feeding, grooming, exercising, or otherwise providing care to promote and maintain the well-being of pets and other animals in settings such as kennels, animal shelters, and zoos.
0	**Applied Arts and Design** — Work involves applying artistic principles to the design and creation of various objects and materials.
0	**Athletics** — Work involves training, practicing, or competing in athletic events, or supporting those participating in athletics through coaching, training, or officiating.
0	**Culinary Art** — Work involves preparing, cooking, and presenting food in a distinctive style usually associated with a specific culture or region.
0	**Life Science** — Work involves studying and researching living organisms through scientific disciplines such as biology, botany, zoology, microbiology, physiology,

Occupational Interest	Specific Interest Area
	biochemistry, or zoology.
0	**Music** — Work involves conducting, composing, or playing music or singing.
0	**Nature/Outdoors** — Work involves planning, restoring, and conserving natural areas and wildlife habitats.
0	**Performing Arts** — Work involves acting as a performer in dance, comedy, or dramatic productions, choreographing of dance performances, or producing of live or recorded entertainment.
0	**Religious Activities** — Work involves conducting religious worship, leading a church or other religious group, and offering moral guidance associated with beliefs and practices of a religious faith.
0	**Visual Arts** — Work involves creating original visual artwork for a variety of media and purposes using various techniques and software.

Work Styles

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Impact	Work Style
89	**Attention to Detail** — A tendency to be detail-oriented, organized, and thorough in completing work.
88	**Dependability** — A tendency to be reliable, responsible, and consistent in meeting work-related obligations.
78	**Cautiousness** — A tendency to be careful, deliberate, and risk-avoidant when making work-related decisions or doing work.
77	**Integrity** — A tendency to be honest and ethical at work.
73	**Intellectual Curiosity** — A tendency to seek out and acquire new work-related knowledge and obtain a deep understanding of work-related subjects.
65	**Adaptability** — A tendency to be open to and comfortable with change, new experiences, or ideas at work.

Impact	Work Style
59	**Perseverance** — A tendency to exhibit determination and resolve to perform or complete tasks in the face of difficult circumstances or obstacles at work.
58	**Achievement Orientation** — A tendency to establish and maintain personally challenging work-related goals, set high work-related standards, and exert high effort toward meeting those goals and standards.
54	**Initiative** — A tendency to be proactive and take on extra responsibilities and tasks that may fall outside of one's required work role.
53	**Stress Tolerance** — A tendency to cope and function effectively in stressful situations at work.
48	**Self-Control** — A tendency to remain calm and composed and to manage emotions effectively in response to criticism or difficult situations at work.
44	**Tolerance for Ambiguity** — A tendency to be comfortable with ambiguity and uncertainty at work.
43	**Innovation** — A tendency to be inventive, to be imaginative, and to adopt new perspectives on ways to accomplish work.
42	**Self-Confidence** — A tendency to believe in one's work-related capabilities and ability to control one's work-related outcomes.
39	**Cooperation** — A tendency to be pleasant, helpful, and willing to assist others at work.
19	**Sincerity** — A tendency to be genuine and sincere in interactions with others at work, without concern for personal gain or self-interest.
18	**Leadership Orientation** — A tendency to lead, take charge, offer opinions, and provide direction at work.
11	**Optimism** — A tendency to exhibit a positive attitude and positive emotions at work, even under difficult circumstances.
11	**Social Orientation** — A tendency to seek out, enjoy, and be energized by social interaction at work.
7	**Humility** — A tendency to be modest and humble when interacting with others at work.

Impact	Work Style
4	**Empathy** — A tendency to show concern for others and be sensitive to others' needs and feelings at work.

Workforce Characteristics

Wages & Employment Trends

Median wages (2025) $47.66 hourly, $99,130 annual

State wages Select a State Go

Local wages ZIP Code Go

Employment (2025) 323,600 employees

Projected growth (2025-2035) Decline (-1% or lower)

Projected job openings (2025-2035) 13,400

State trends Select a State Go

Top industries (2025) Professional, Scientific, and Technical Services (25% employed in this sector)
Educational Services (13%)

Source: Bureau of Labor Statistics 2025 wage data and 2025-2035 employment projections. "Projected growth" represents the estimated change in total employment over the projections period (2025-2035). "Projected job openings" represent openings due to growth and replacement.

Job Openings on the Web

State job openings Select a State Go

Local job openings ZIP Code Go

More Information

Related Occupations

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- 15-1241.00 Computer Network Architects Bright Outlook
- 15-1231.00 Computer Network Support Specialists
- 15-1211.00 Computer Systems Analysts
- 15-1299.08 Computer Systems Engineers/Architects
- 15-1232.00 Computer User Support Specialists
- 15-1242.00 Database Administrators
- 15-1243.00 Database Architects
- 15-1212.00 Information Security Analysts
- 15-1299.05 Information Security Engineers
- 15-1252.00 Software Developers

Supplemental related occupations

- 15-1241.01 Telecommunications Engineering Specialists
- 15-1299.01 Web Administrators
- 11-3021.00 Computer and Information Systems Managers
- 15-1299.04 Penetration Testers
- 15-1253.00 Software Quality Assurance Analysts and Testers
- 15-1251.00 Computer Programmers
- 49-2022.00 Telecommunications Equipment Installers and Repairers, Except Line Installers
- 15-1299.07 Blockchain Engineers
- 15-1299.09 Information Technology Project Managers
- 15-1254.00 Web Developers

Professional Associations

Disclaimer: Sources are listed to provide additional information on related jobs, specialties, and/or industries. Links to non-DOL Internet sites are provided for your convenience and do not constitute an endorsement.

National Associations

All 6 displayed

- American Library Association
- Association for Computing Machinery
- Healthcare Information and Management Systems Society
- IEEE Computer Society
- National Center for Women and Information Technology
- USENIX

Accreditation, Certification, & Unions

- CompTIA ALLY
- Building Industry Consulting Service International